



Syntax I

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Please pick up your **name card** from the front

Extra Credit

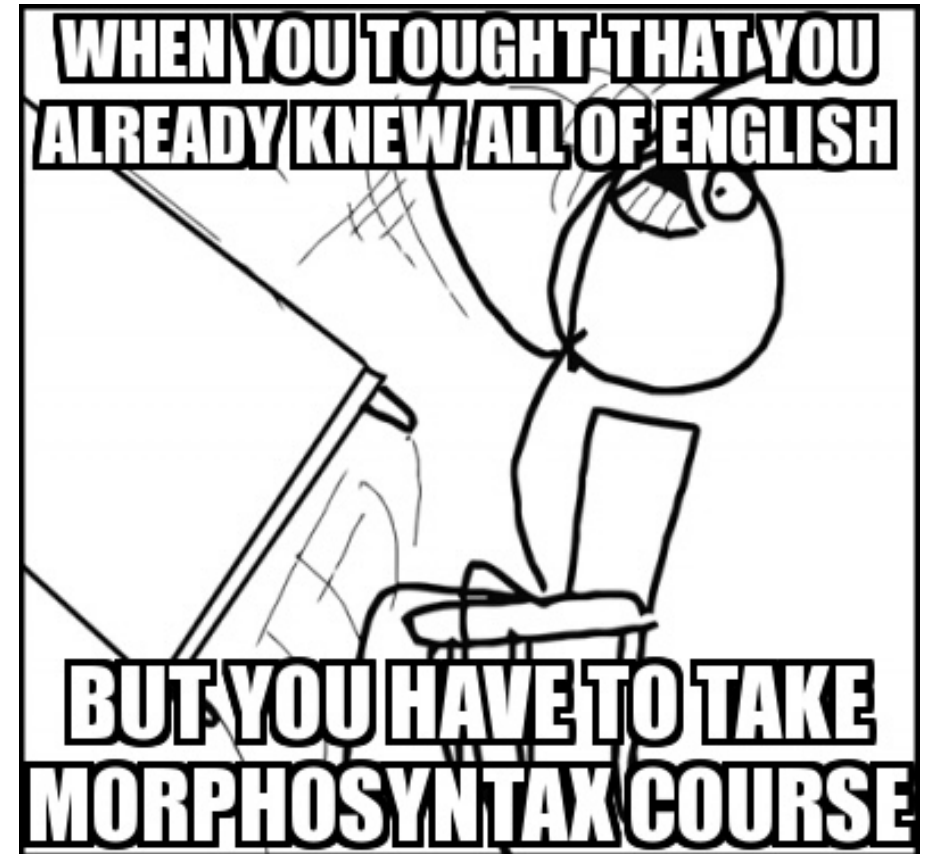
- up to **2%** of extra credit can be earned by participating in **research studies** conducted by members of the Linguistics Department
- these studies usually consist of experiments that take 25–45 minutes
- experiments are offered through the Linguistics department experiment management system (SONA)

SONA MUST KNOWS

- your SONA account will be created for you **automatically** and you **should not** have to create or request your own account
- your login information will be sent to the email address listed on the roster from the registrar
- you are responsible for ensuring the accuracy of your course assignments in SONA
- you are able to fix your own course assignments in the **SONA system**
- you are responsible for making sure your credit assignments are allocated to the right courses
- you can check out the SONA **tutorial** at this link, under “Labs”, click on “Experiment System Tutorial”
 - <https://ling.rutgers.edu/research-mainmenu/research-labs-mainmenu-145>

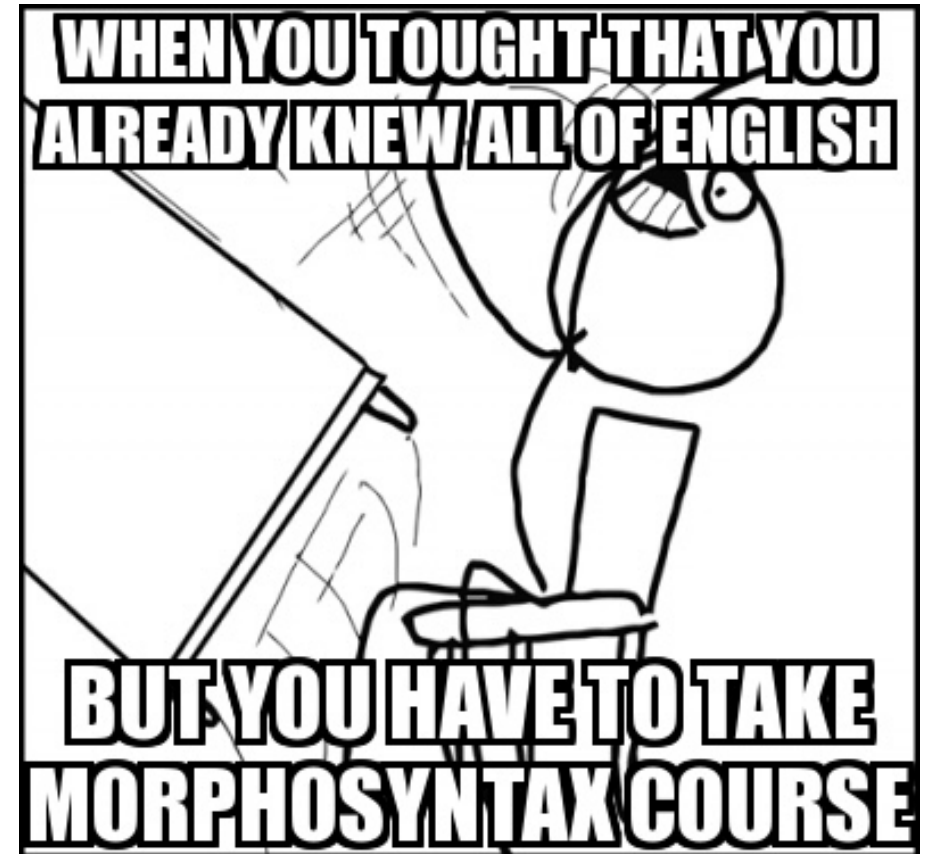
Introduction to Syntax

What is the **basic** unit in Morphology?



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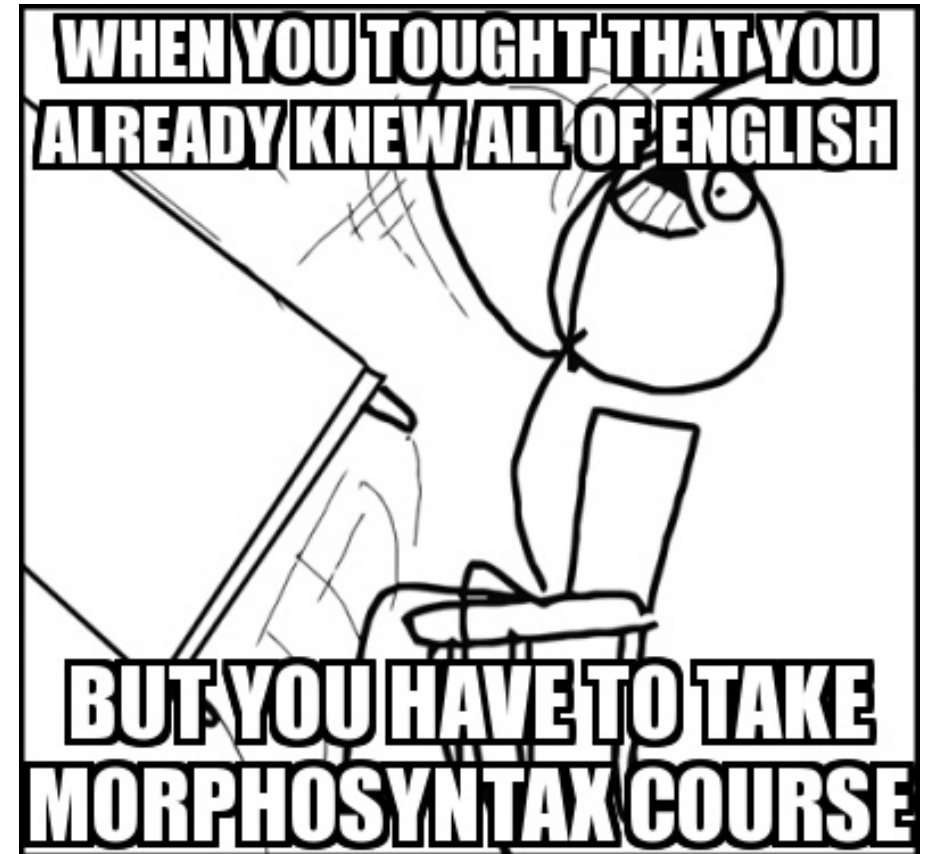
MORPHEME



What is the **basic** unit in Morphology?

MORPHEME

What happens if **larger** units were built?

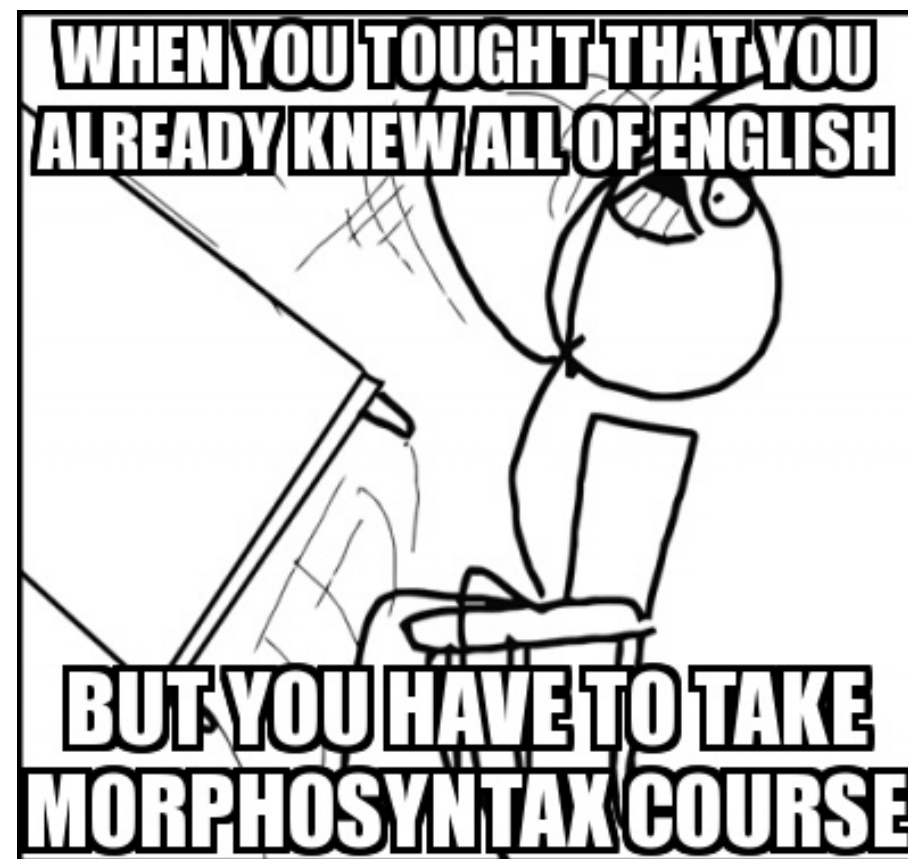


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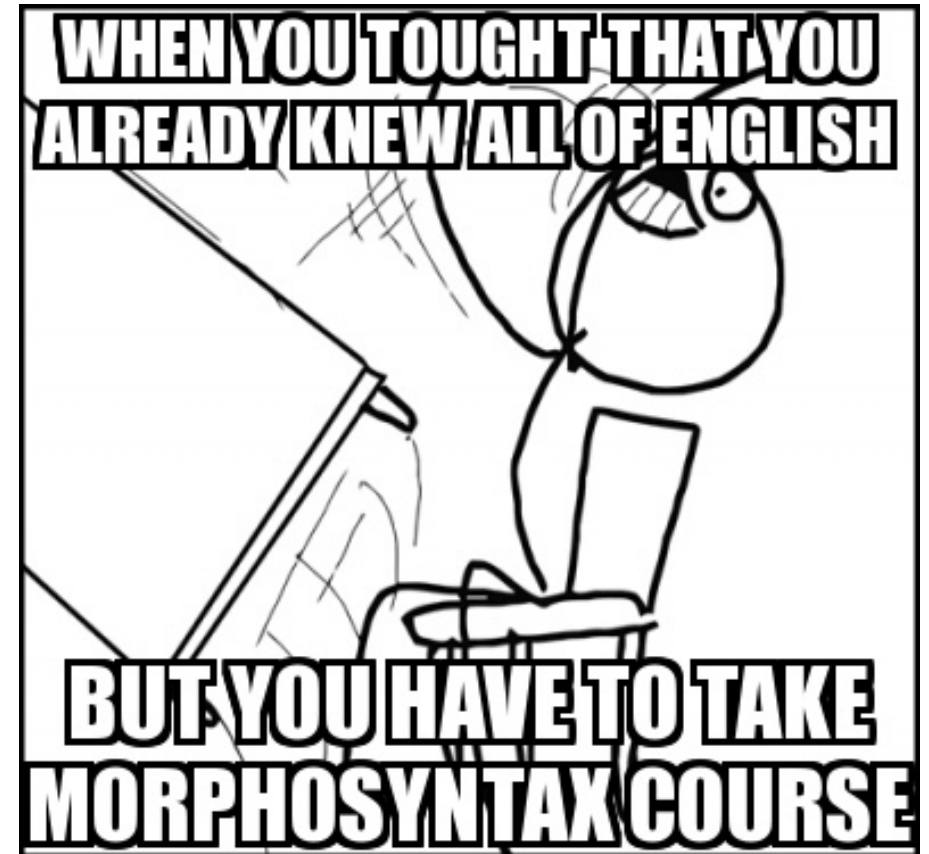


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SYNTAX



Morphology cares about word structure

Syntax cares about sentence structure

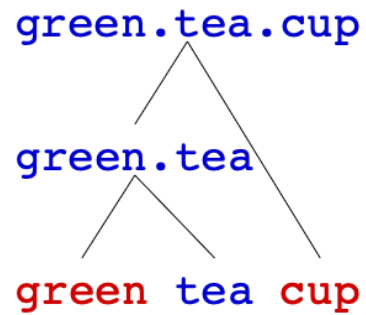
Moving onto More Ambiguous Cases

un-lockable vs. unlock-**able**

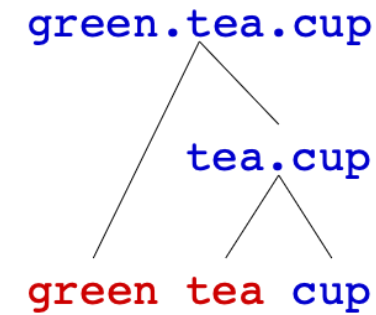
green tea cup

Moving onto More Ambiguous Cases

green tea cup



green tea cup



Exercise I: Structural Ambiguity

German history professor

German history professor

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German history professor

German.history.professor

German.history

German history professor

German history professor

german.history.professor

history.professor

German history professor

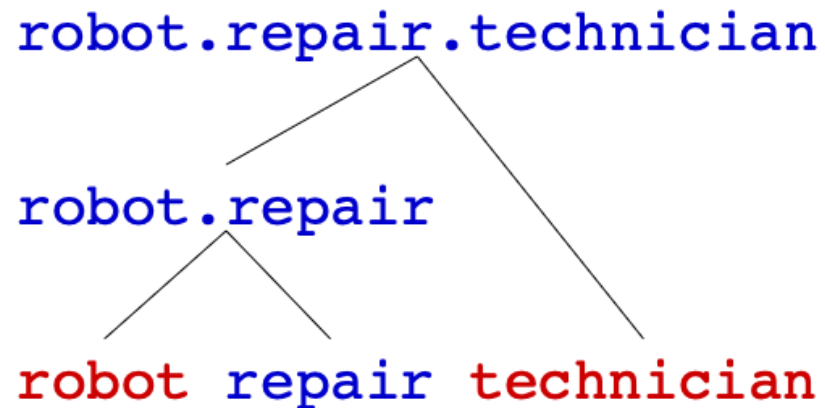
Exercise I: Structural Ambiguity

robot repair technician

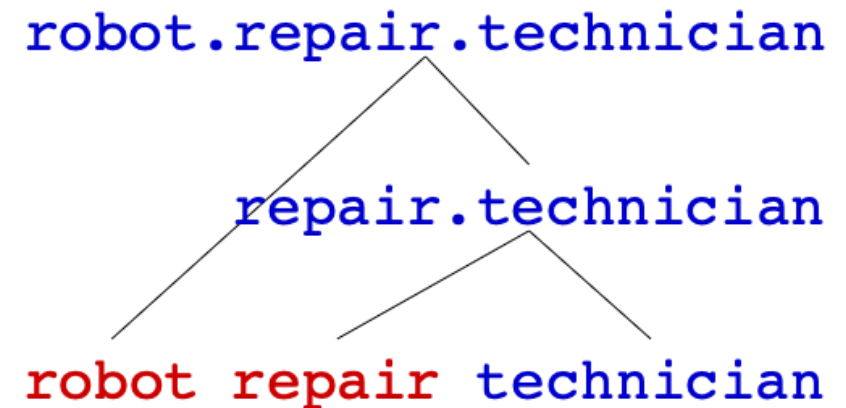
robot repair technician

Exercise I: Structural Ambiguity

robot repair technician



robot repair technician



Lexical Ambiguity

Lexical Ambiguity

The presence of two or more possible meanings within a single word.



"I saw her duck."

ThoughtCo.

[Chinese > English] It's supposed to say Cancer in traditional Chinese, does it?

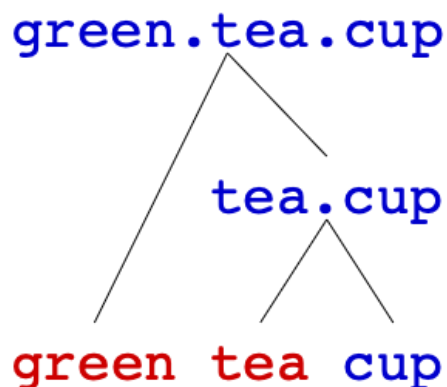


Structural Ambiguity

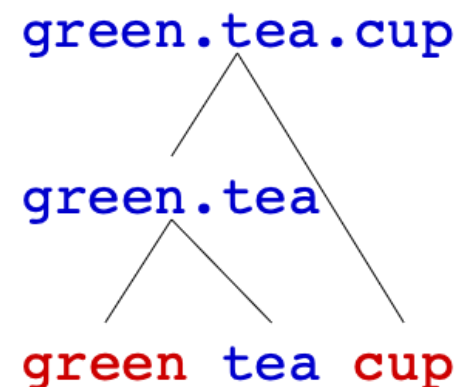
words (or phrases and sentences) are structured **hierarchically**, not linearly

- within a word, the same **morpheme order** can sometimes have **different** word meanings
 - *un-lockable* vs. *unlock-able*
- within a phrase or a sentence, the same **word order** can sometimes have different meanings depending on the different structures

[green [tea cup]]



[[green tea] cup]



Structural Ambiguity for Sentences

sentences sometimes can be **ambiguous** depending on different hierarchical structures even when each word in the sentence has exactly **the same meaning** and they are put in **the same order**

I saw the man with a telescope



I used a telescope to see the man



I saw the man who had a telescope

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(1) *Arthur read the book on the desk.*

Arthur *read the book on the desk*

Arthur *read the book on the desk*

(2) *They helped the student with homework.*

Structural Ambiguity for Sentences

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Arthur *read* *the book* *on the desk*

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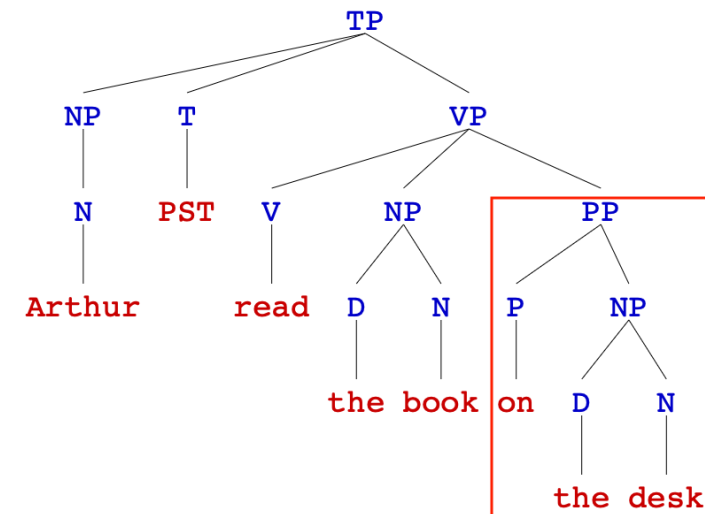
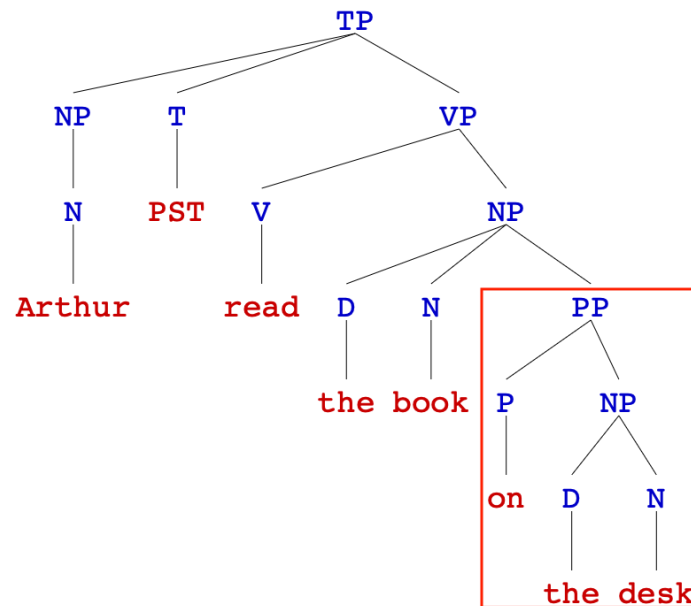
(2) *They helped the student with homework.*

They *helped* *the student* *with homework*

They *helped the student with homework*

Structural Ambiguity for Sentences

Arthur read the book on the desk.



do not worry about the **trees** for now, just check what *on the desk* is attaching to

Syntax is the aspect of our competence that deals with **phrase and sentence structure**.

Word Order: Variation

English: SVO

Mary
S

loves
V

the cat
O

* *loves Mary the cat.*

Word Order: Variation

English: SVO

Mary
S

loves
V

the cat
O

* loves Mary the cat.

Irish: VSO

bhuail
hit
V

sé
he
S

mé
me
O

'He hit me.'

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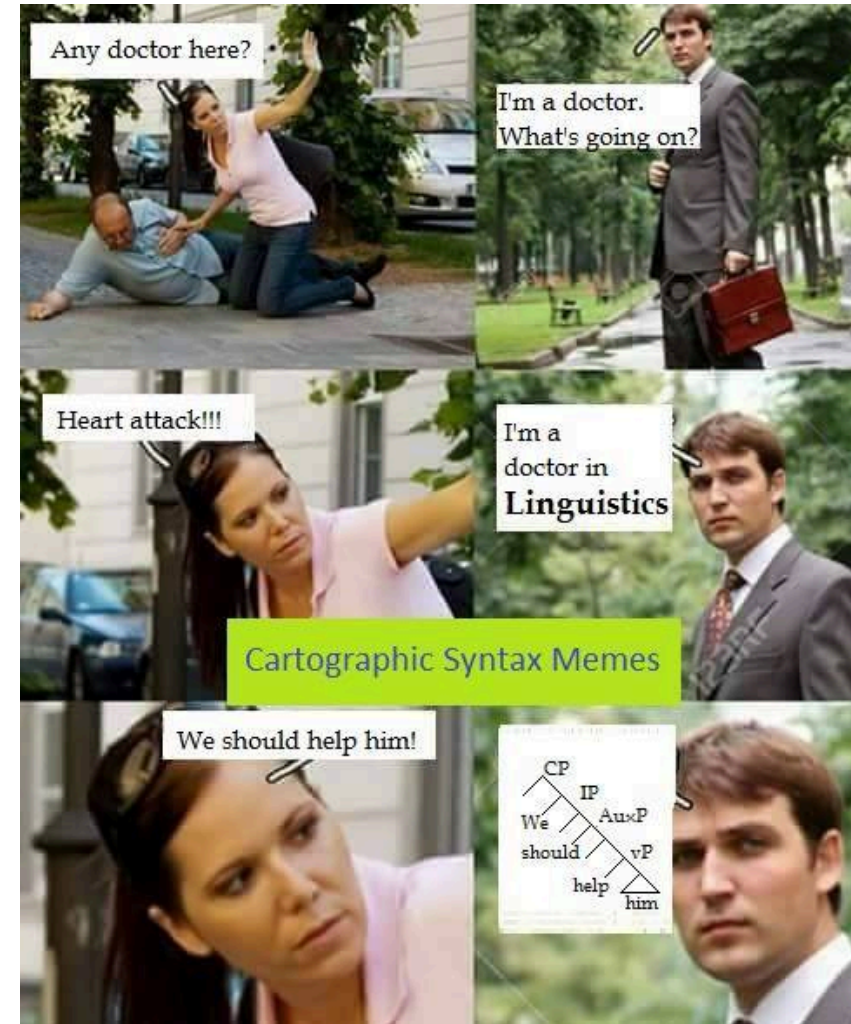
ek seb
an apple
O

chaahiye
want
V

'I want an apple.'

Structure

- **word order** is one relevant aspect of how sentences are put together
- but that's **NOT** enough:
 - *Arthur read the book on the table.*
 - *Arthur read the book on the table.*
- the same string of words has two different meanings
- **hierarchical structure** is also relevant for putting words together



Part of Speech

From the Ground Up

large phrases and sentences



phrases



words



morphemes

From the Ground Up

large phrases and sentences



phrases



words



morphemes

- like morphology, we'll build our theory of syntax **ground-up**
- instead of starting with morphemes, we will start with **words**
- **first task**: figure out what **part of speech** (a.k.a syntactic category, word class) a word has
- once we have got that down, we'll see how they **combine** to form phrases
- notice the parallels with Morphology, we start from each **morpheme**, then we **combine**

Parts of Speech

- parts of speech (aka lexical classes) are categories we use to **classify words**:
 - nouns, verbs, adjectives, adverbs...
- they're important for **Morphology**:
 - **derivation** can change a word's part of speech
courage (**N**) → *en-courage* (**V**) → *en-courage-ment* (**N**)
 - **inflection** is specific to a part of speech: nouns, verbs, and adjectives all inflect differently
walk → *walk-ed* (-ed for **V** only)
- they'll also be important for **Syntax**:
 - sentences are made of words, and a word's part of speech determines its **role** in the sentence:
 - for instance, the subject of a sentence is always a noun
every sentence needs a main verb...

Parts of Speech

- for following words, it is easy to define what they are in terms of **part of speech**

noun

denotes a person, place, thing

verb

denotes actions

adjective

adds a description to a noun

Parts of Speech

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- so we can provide definitions of these categories, because we know the **meaning** of the word
 - table is a noun because we know it names a thing
- can we figure out a word's part of speech without knowing what it means – just by seeing how it's used in a sentence?
 - the answer is **YES**
 - but the part of speech of a word, is determined by its **distribution**, NOT by its meaning
- **syntactic distribution**: different categories of words appears in different positions in a sentence
- **morphological distribution**: different categories of words takes different affixes

Jabberwocky

- the words are nonsense, but the **syntax** is recognizably English
- **meaning** does not tell us anything now, we have to rely on how they're being used
- **step 1**: on the form, take your best guess at the part of speech of each **coloured** word (noun, verb, adj)
- **step 2**: compare your answers with that of a neighbor
 - did you get the same answers?
 - what made you choose the answer you did?

Tw'as brillig, and the **slithy toves**

Did **gyre** and gimble in the **wabe**;

All mimsy were the **borogroves**,

And the mome **raths outgrabe**

Lewis Carroll, *Jabberwocky* (1871)



Morphological Distribution

for **morphological distribution**, we check:

- what morphemes does it have?
- are they associated with a specific part of speech?
- consider the word **borogroves**:
 - it seems to end in the plural suffix **-s**
 - what other words use plural -s?
- we conclude **borogroves** is a noun (**N**)
 - because it fits the **morphological distribution** of nouns
 - it uses morphemes associated with nouns

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Syntactic Distribution

for **syntactic distribution**, we check:

- where does the word appear in the sentence?
- what other words appear around it?
- consider the word **borogroves** once more:
 - it comes after **the** → *the borogoves*
 - what other words can follow **the**?
- we still conclude that **borogroves** is a noun (**N**)
 - because it fits the **syntactic distribution** of nouns
 - it goes where nouns go in the sentence

Twas brillig, and the **slithy toves**

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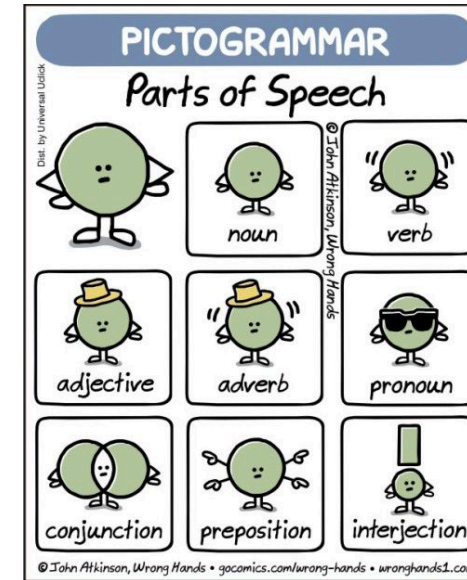
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Part of Speech

parts of speech aren't just random categories you have to memorize in English class

- words of a particular part of speech tend to behave in **similar** ways:
 - they have a similar **morphological distribution** (similar morphemes)
 - and a similar **syntactic distribution** (similar role in sentences)
-
- that is, they follow a **pattern**
 - a **pattern** that forms part of our **mental grammar**
 - we pick up these **patterns** when acquiring our native language
 - a small child might not know what a **noun** is, but they might know *take ball, take toy...* but not * *take eat*



Parts of Speech: Replacement Test

since these categories are part of our **mental grammar**, they give us intuitions about how a word **can** or **cannot** be used in a sentence

- we can use these intuitions in a **replacement test** to figure out a word's part of speech:
- let's say for the following sentence, we do not know the part of speech of **delicious**:
 - *John ate a **delicious** sandwich.*
- but we know the part of speech of **eat** (V) and **good** (Adj)
- we replace **delicious** with either *eat* or *good*
 - * *John ate a(n) **eat** sandwich.*
 - *John ate a **good** sandwich.*
- if the sentence after replacing is still **well-formed**, *delicious* probably has the same part of speech with the replaced word

Parts of Speech: Replacement Test

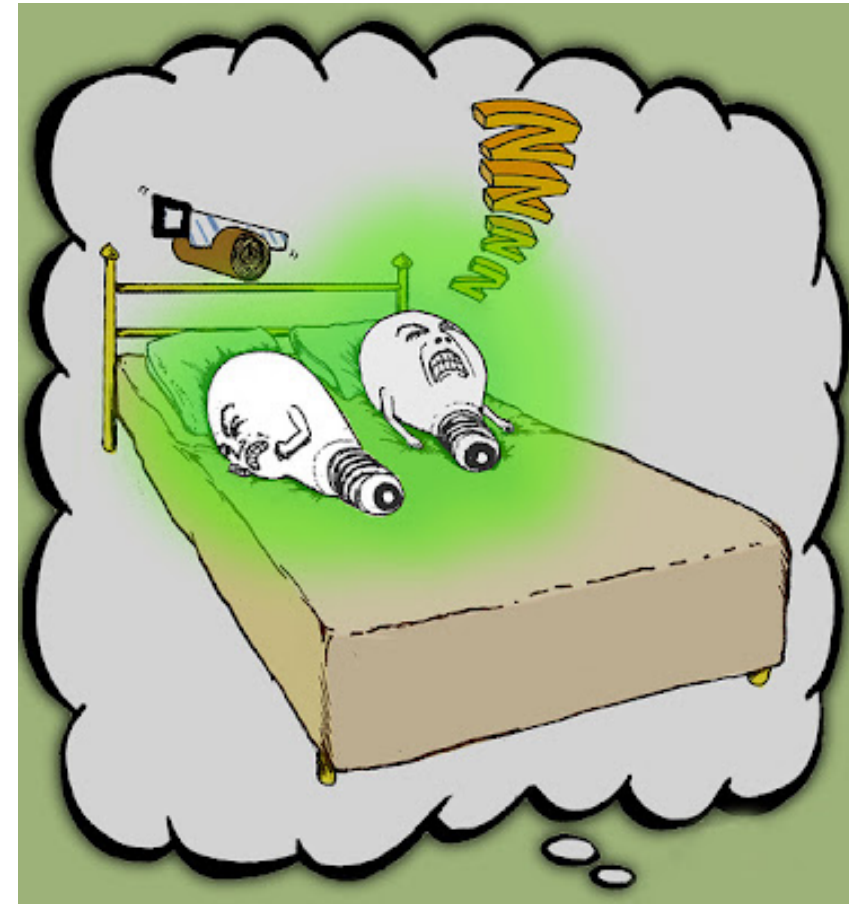
note here it might NOT **always** work!

let's say we know the part of speech of the word **friendly** (Adj) and **good** (Adj)

- (a) ? *John ate a **friendly** sandwich.*
- (b) * *John ate a(n) **eat** sandwich.*
- (c) *John ate a **good** sandwich.*

comparing the first two sentences, we know that

- **(a)** is syntactically well-formed, but not meaningwise proper (**semantic anomaly**)
- but **(b)** is syntactically ill-formed (**syntactic ungrammaticality**)



Colorless green ideas sleep furiously.

we will introduce following parts of speech and their **morphological** and **syntactic distributions**

noun (N): *dog, cat, teacher*

verb (V): *eat, do, think*

adjective (Adj): *easy, happy, wonderful*

adverb (adv): *easily, happily, wonderfully*

determiner (D): *the, my, a*

preposition (P): *in, with, to₁*

conjunction (Conj): *and, or, because*

complementiser (C): *that, whether*

tense (T): *be, will, to₂*

Nouns

nouns denote someone or something

- **nouns** denote a person, place, or thing...
- also abstract entities, concepts, even actions

student, school, desk

hope, idea, year, math fishing

nouns can be further divided into

- **common nouns**
- **proper nouns** (names)

girl, river, college

Mary, Raritan, Rutgers

Nouns

morphological distribution: what affixes can a **noun** have in English?

- nouns can take plural **-s** or possessive **'s**
- derivational morphemes that form nouns: -
tion/sion, **-ness**, **-ity**...

cat**s**, dog**s**, Merlin'**s**, student**'s**

happi**ness**, read**ing**, encouragem**ent**

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syntactic distribution: where do **nouns** appear in sentences in English?

- nouns can follow **determiners** like *a, the, my* **a** dog, **the** cat, **my** friend
- nouns can follow **adjectives** **friendly** dog, **angry** cat, **best** friend
- nouns can follow **prepositions** **at** school, **in** line, **on** demand
- nouns can be **subjects** of sentences **Dogs** are humans' best friend.

nouns can be **objects** of sentences

I saw **dogs**.

Pronouns

- **pronoun** takes the place of a noun
 - * Mary had a little lamb = **She** had it.
- some linguists consider pronouns a **subcategory** of nouns, because they behave like nouns in many ways:
 - they can be **subjects**:
 - Merlin sings = **He** sings.
 - they can be **objects**:
 - Did you help your mom? = Did you help **her**?
- some linguists consider pronouns a **separate** part of speech, because in some ways they behave differently:
 - they (usually) aren't modified by
 - the** or an **adjective**
 - although: lucky **you**! (idiomatic use)
 - the happy student
 - * the **them**
 - * happy **them**
- in this class, we'll consider them a **subcategory** of nouns, but also acknowledge their differences

Verbs

verbs can denote

- **actions** sit, jump, fly
- **states** be, have, know, want
- or **processes** dream, feel, rest



Verbs

morphological distribution: what kind of morphemes can **English verbs** have?

- verbs can be conjugated for **tense** & **subject**

I break, she break**s**, we **broke**, we have broken

- derivational morphemes can make verbs
-ise, -ify, -ate, -en...

modern**ise**, elect**ify**, valid**ate**, length**en**

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syntactic distribution: where do **verbs** go in **English** sentences?

- verbs go **after** the subject and **before** object(s)

She **knows** me, I **did** it.

- all sentences have at least **one verb**

I **like** pizza. * I an ice cream.

- some sentences have **auxiliary** verbs in addition to the **main** verb

I **am** singing. I **have** finished.

- modal** verbs are a type of auxiliary verb that express obligation, possibility, etc.

can, could, may, might, shall, should, would, must, ought to

Adjectives and Adverbs

adjectives and **adverbs** modify other words
providing additional description

- **adjectives** modify **nouns**
- **adverbs** modify **verbs**
adjectives and
other **adverbs**

ferocious chihuahua

The chihuahua barked **angrily**.
It was **extremely** aggressive.
It ran **very** quickly.

syntactic distribution

- both go **before** the word they modify:
- but **adverbs** can go **before** or **after** verbs

playful labrador, **very** friendly

I **quickly** stopped. / I stopped **quickly**.

Adjectives and Adverbs

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providing additional description

morphological distribution

- many English adjectives can inflect for **comparative** and **superlative** with *-er*, *-est*
- **derivational** morphemes for adjectives *-ish*, *-al*, *-ic*...
- **derivational** morpheme for adverbs: *-ly*
 - though not all adverbs end in *-ly*
 - and some adjectives end in *-ly*, too

nice**er**, nice**est**

* perfect**er**, * perfect**est**

book**ish**, person**al**, horrific**ic**

happi**ly**, slow**ly**

My dog runs **quickly**

the friend**ly** cat, the kind**ly** man

Determiners

determiners help to specify a word's reference

determiners gather together many words you may have learned as separate categories:

- articles *the, a*
- possessive pronouns *my, our, your, his, her, their, its*
- demonstratives **this** apple, **that** barn
- interrogative determiners **which** computer, **what** floor
- quantifiers *all, some, many*

determiners help to specify a word's reference, what **actual** thing in the world a word refers to

- dog vs. **my** dog

Determiners

morphological distribution

- English determiners don't have any specific morphemes that are typical of them
- in fact, they tend to be **monomorphemic**

syntactic distribution

- determiners can go **before** a noun
- or **before** an adjective, **then** a noun
- English usually doesn't allow **two** determiners in a row

the apple, **my** friend

the red apple, **my** good friend

* **the my** friend

Prepositions

prepositions indicate some relationship between a **noun** and the **other words** in a sentence

- **prepositions** can indicate:

- **spatial** relationship

on the table

- **grammatical** relationships

belong **to** me

- **conceptual** relationships

in the morning

morphological distribution: no specific affixes and are often **monomorphemic**

syntactic distribution

- prepositions go **before** nouns in English

on topic, **in** time, **with** cheese

(pre position)

Conjunctions and Complementisers

conjunctions join together two words, phrases, or clauses

- *and, or, but, because, if, since*
- **coordinating conjunctions** join two words/phrases of the **same type**
 - *and, or, but*
 - *cats **and** dogs, coffee **or** tea*
 - **syntactic distribution**: between the two elements being joined
- **subordinating conjunctions** are used to insert one sentence (subordinate clause) into another sentence (main clause) and they are also called **complementisers**
 - *because, that, if, since, whether*
 - *I went to bed **because** I was tired. / **If** you study, you will succeed.*
 - **syntactic distribution**: before the subordinate clause being inserted

Tense and Aspect

tense refers to when an action occurs
whether it happens in the **past**, **present** or **future**

- tells **WHEN** the action occurs
- **past** tense: *She walk**ed** to school.*
- **present** tense: *She walk**s** to school*
- **future** tense: *She **will** walk to school.*

PST
PRES
FUT

aspect refers to the state of the action
whether it is **ongoing**, **completed** or **habitual**

- tells **HOW** the action unfolds
- **progressive** aspect: *She **is** walk**ing** to school.*
- **perfective** aspect: *She **has** walked to school.*
- **habitual** aspect: *She walk**s every day**.*

PROG
PERF
HAB

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auxiliaries

has/have/had, am/is/are/was/were, do/does/did

modals

*will, would, shall, should,
can, could, may, might, must*

non-finite tense marker

to

negation (Neg): not

Example from another Language

Nootka, Wakashan

(a) mamu:k-**ma**
 working-**PRES**
 'The man is working.'

qu:ʔas-ʔi
 man-**DEF**

- **PRES** = present tense
- **DEF** = definite

(b) qu:ʔas-**ma**
 man-**PRES**
 'The working one is a man.'

mamu:k-ʔi
 walking-**DEF**

1. What's the **noun** in (a)? What's the **verb**?
2. What's the **noun** in (b)? What's the **verb**?

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- **DEF** = definite

(b) qu:ʔas-**ma**
 man-**PRES**
 'The working one is a man.'

mamu:k-ʔi
 walking-**DEF**

1. What's the **noun** in (a)? What's the **verb**?
2. What's the **noun** in (b)? What's the **verb**?

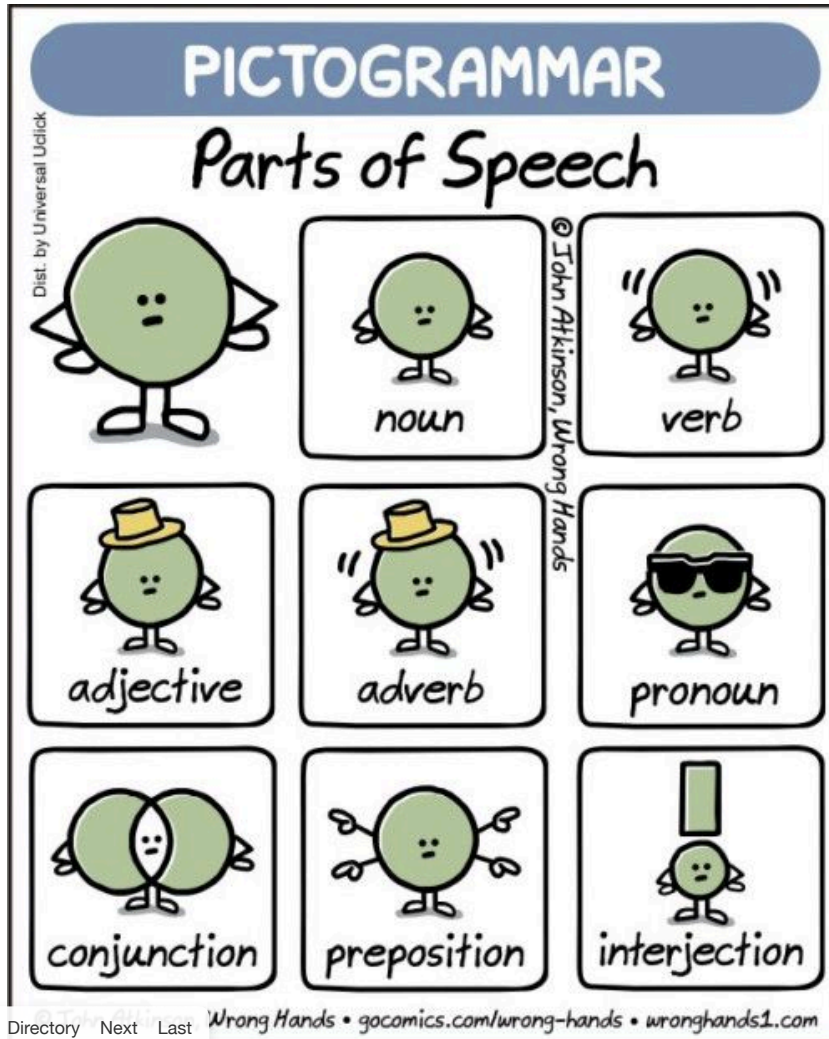
in both sentences:

- the first part is the **verb**
 - indicated by the presence of the **present tense** morpheme **-ma**
- the second part is the **noun**
 - indicated by the **definite** morpheme **-ʔi**

Open and Closed Classes

- parts of speech can be divided into **open** and **closed classes**
- **open class**: words have more of a **lexical** meaning
 - typical dictionary definition
 - it's fairly easy to *borrow* or *coin* new words in these categories
 - nouns, verbs, adjectives, and adverbs
- **closed class**: words have more of a **grammatical** meaning
 - it's less common, but not impossible, to borrow or coin new words in these categories
 - determiners, prepositions, conjunctions, complementisers, tense

Exercise II: Parts of Speech



for every word in the sentences below, identify its **parts of speech** (lexical category)

1. Mary will come tomorrow.
2. John went to New York yesterday.
3. A cat is under the table.
4. Her dad was fixing a car.

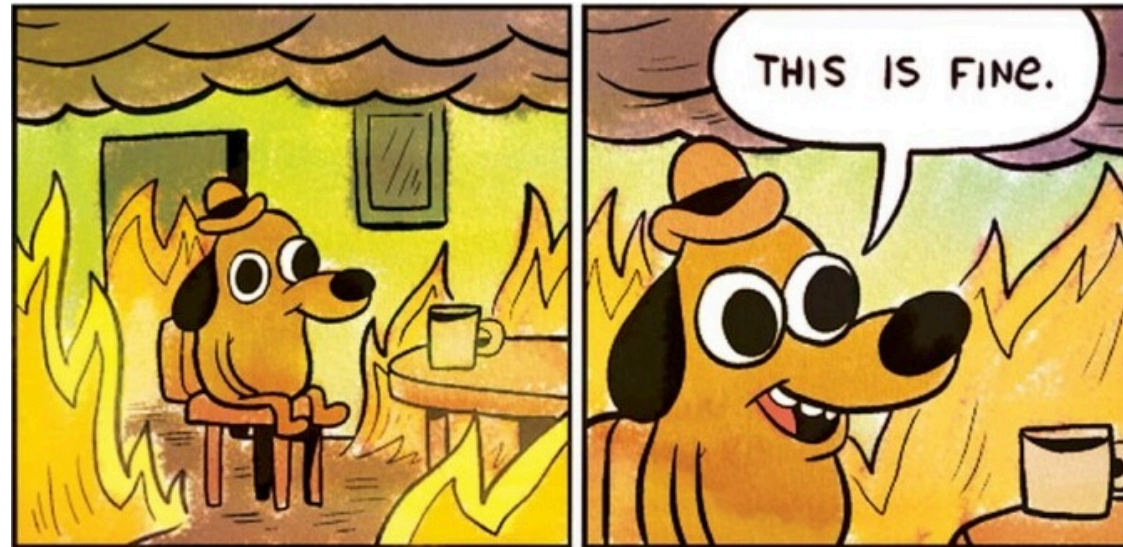
Summary

- **parts of speech** are categories of words that tend to behave in similar ways:
 - noun, verb, adjective, adverb, determiner, preposition, conjunction, complementiser, tense
- we often learn them in school by definition
- but we figure them out here based on **context**, by considering their:
 - **morphological distribution** – what kinds of morphemes they have.
 - **syntactic distribution** – where they tend to appear in the sentence
- by examining their **distribution**, we can notice patterns in the way certain types of words behave
- and this will be fundamental to our understanding of **syntax**, as we start putting words together to form **sentences**

Homework I is due this Sunday (**Feb 15**)

Homework II will be published this Sunday, and it is due next Sunday (**Feb 22**)

continue reading Chapter 2: Part of Speech from *Andrew Carnie's Syntax textbook*



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